

EX NO : 3

WIRESHARK

DATE :

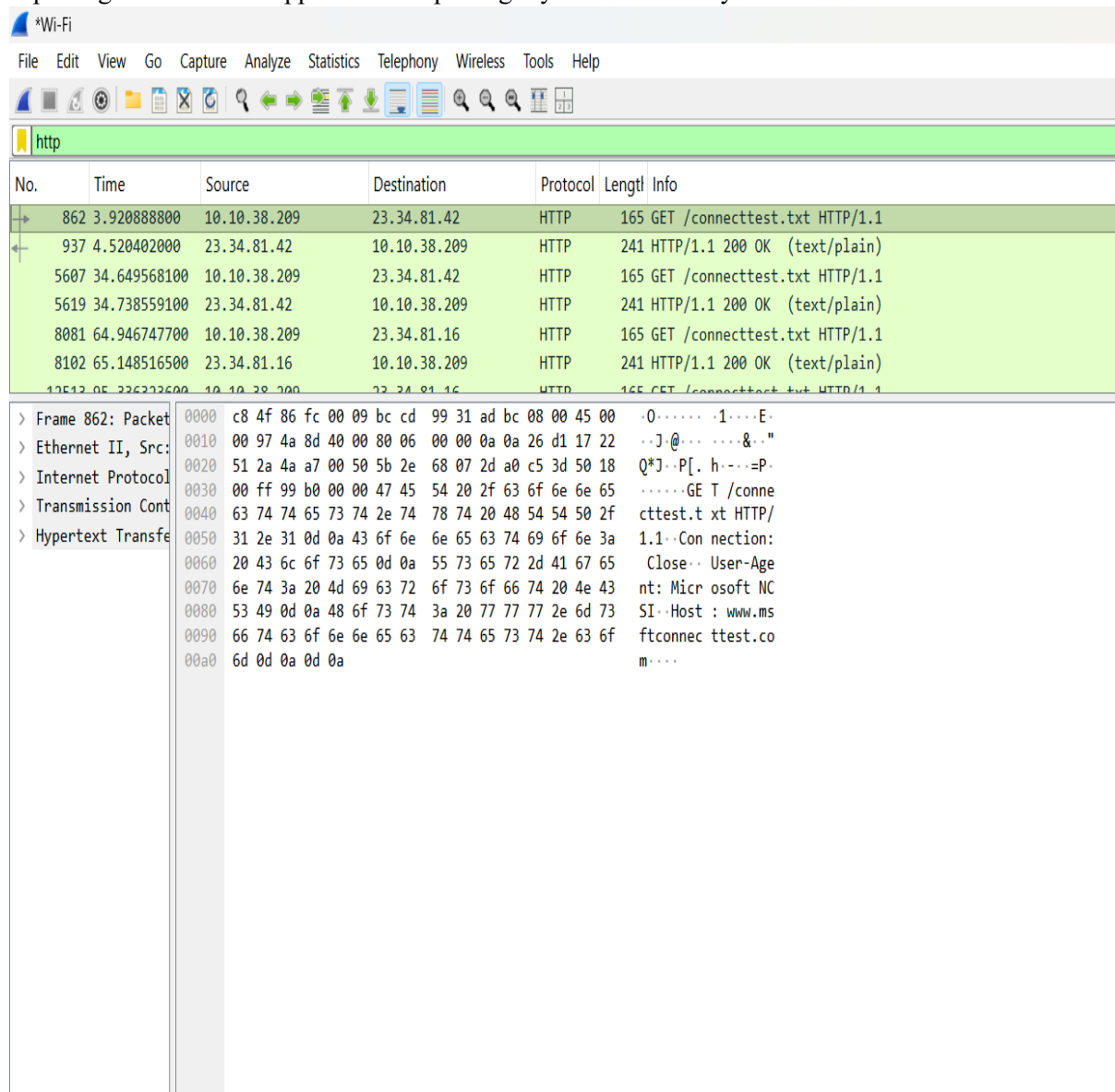
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.

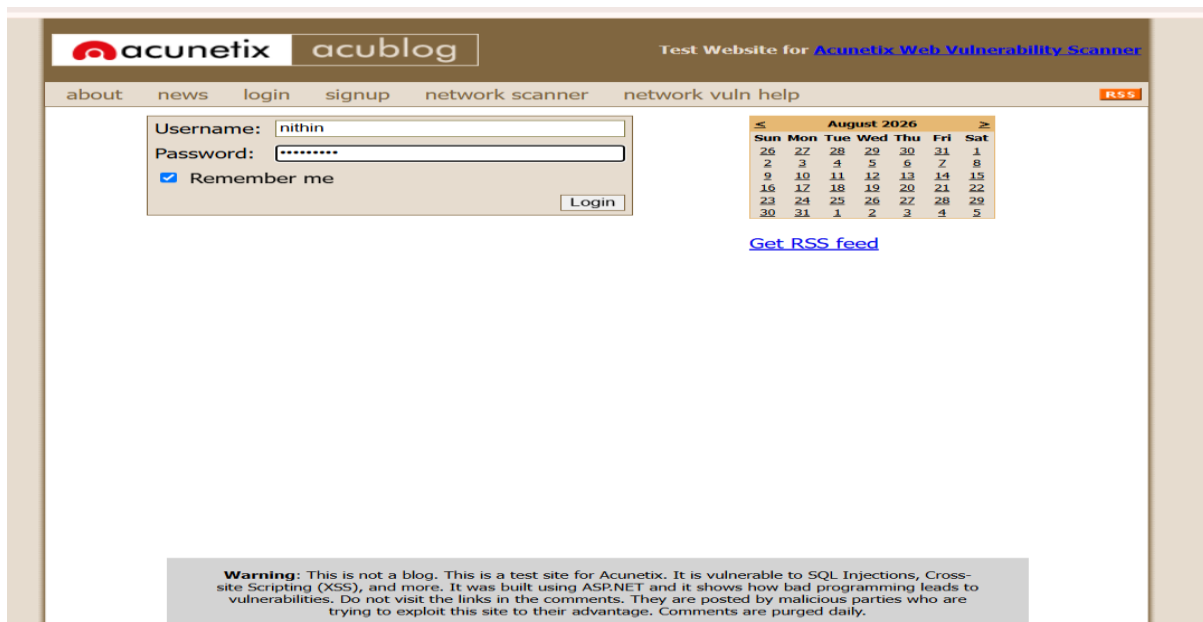


The screenshot shows the Wireshark network capture analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The packet list pane shows a filter set to 'http' and a list of captured packets. The packet details pane for frame 862 is expanded, showing the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
862	3.92088800	10.10.38.209	23.34.81.42	HTTP	165	GET /connecttest.txt HTTP/1.1
937	4.520402000	23.34.81.42	10.10.38.209	HTTP	241	HTTP/1.1 200 OK (text/plain)
5607	34.649568100	10.10.38.209	23.34.81.42	HTTP	165	GET /connecttest.txt HTTP/1.1
5619	34.738559100	23.34.81.42	10.10.38.209	HTTP	241	HTTP/1.1 200 OK (text/plain)
8081	64.946747700	10.10.38.209	23.34.81.16	HTTP	165	GET /connecttest.txt HTTP/1.1
8102	65.148516500	23.34.81.16	10.10.38.209	HTTP	241	HTTP/1.1 200 OK (text/plain)
13512	05.226232600	10.10.38.209	23.34.81.16	HTTP	165	GET /connecttest.txt HTTP/1.1

Frame 862: Packet 0000 c8 4f 86 fc 00 09 bc cd 99 31 ad bc 08 00 45 00 .0.....1....E.
Ethernet II, Src: 0010 00 97 4a 8d 40 00 80 06 00 00 0a 0a 26 d1 17 22 ..J.@...&.."
Internet Protocol 0020 51 2a 4a a7 00 50 5b 2e 68 07 2d a0 c5 3d 50 18 Q*J...P[.h...=P.
Transmission Control 0030 00 ff 99 b0 00 00 47 45 54 20 2f 63 6f 6e 6e 65GE T /conne
Hypertext Transfer 0040 63 74 74 65 73 74 2e 74 78 74 20 48 54 50 2f cttest.t xt HTTP/
0050 31 2e 31 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 1.1..Con nection:
0060 20 43 6c 6f 73 65 0d 0a 55 73 65 72 2d 41 67 65 Close.. User-Age
0070 6e 74 3a 20 4d 69 63 72 6f 73 6f 66 74 20 4e 43 nt: Micr osoft NC
0080 53 49 0d 0a 48 6f 73 74 3a 20 77 77 77 2e 6d 73 SI..Host : www.ms
0090 66 74 63 6f 6e 6e 65 63 74 74 65 73 74 2e 63 6f ftconnec ttest.co
00a0 6d 0d 0a 0d 0a m....

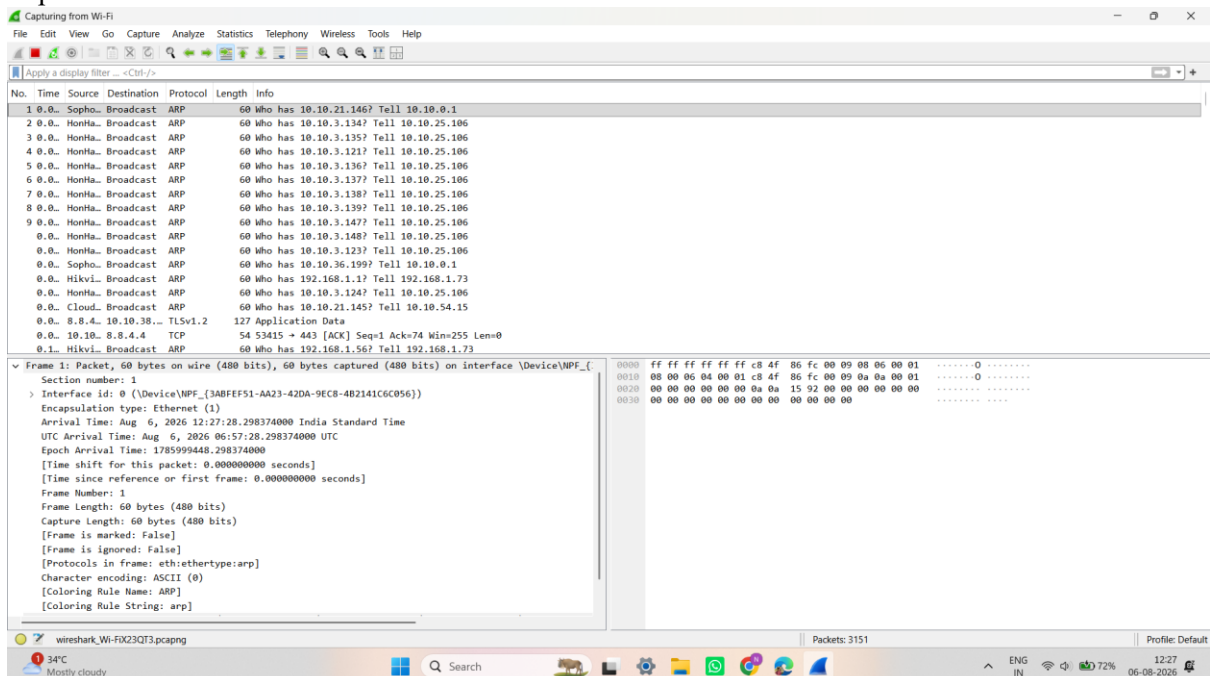
Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image.



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can see in the below image the green bar where we apply the filter.

http



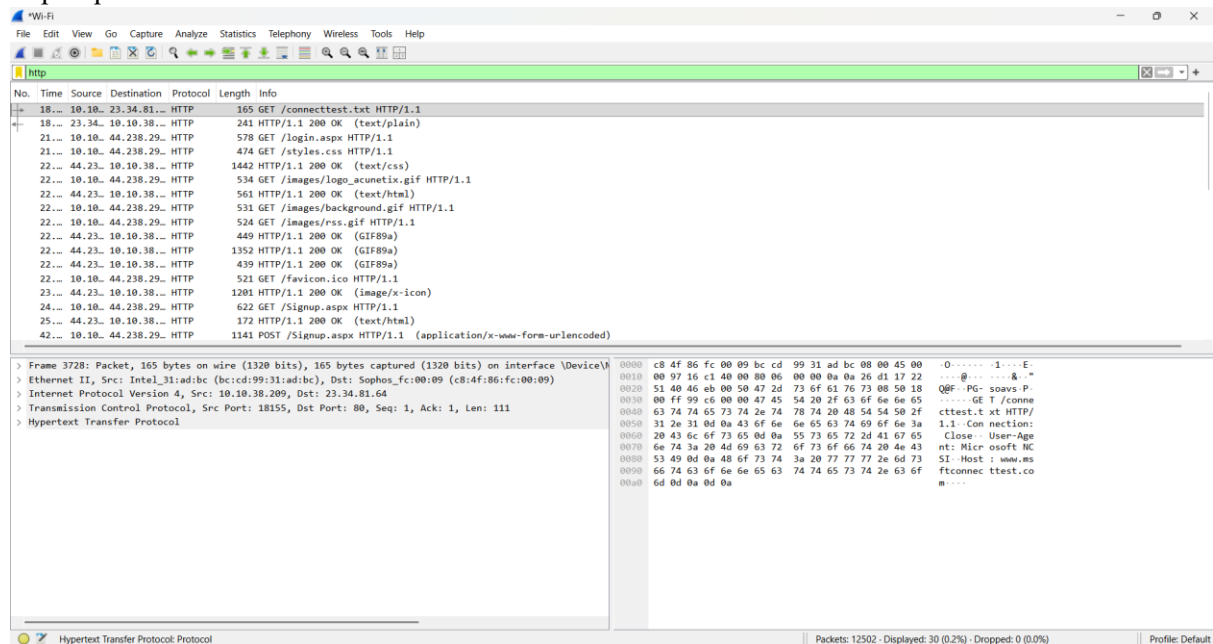
Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-

- GET
- POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

`http.request.method == "GET"`

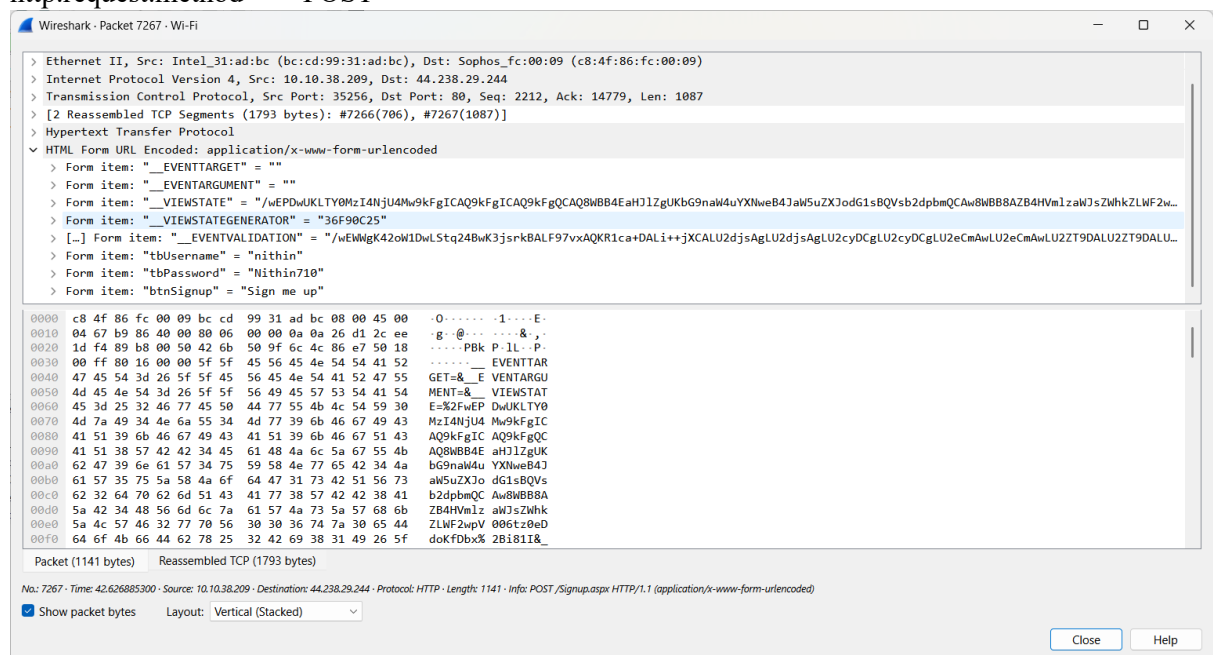


GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

`http.request.method == "POST"`



As you can see we have a packet with form data click on the packet with user info and the application URL encoded, and click on the down-

HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "uname" = "**nithin**"

Form item: "pass" = "**Nithin710**"